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The due date for submitting this assignment has passed.

Due on 2025-10-01, 23:59 IST.

Assignment submitted on 2025-09-30, 16:43 IST

In digital image processing, quantization is a technique used to reduce the number of possible pixel intensity values. This is often done to compress an image or simplify its representation for analysis.

The following Python program demonstrates **intensity quantization** using the **Python Imaging Library** (PIL) and **NumPy**. The program loads a grayscale image, examines the intensity of each pixel, and replaces it with a new value based on which range it falls into. The intensity values are divided into 8 equal levels ranging from 0 to 255.

```
from PIL import Image
import numpy

# Step 1: Load the image in grayscale mode
im = Image.open('images.jpg').convert('L')  # Ensure grayscale
pixelMap = im.load()

# Step 2: Create a new grayscale image
img = Image.new('L', im.size)
pixelNew = img.load()

# Step 3: Quantize
for i in range(img.size[0]):
    for j in range(img.size[1]):
        if 0 <= pixelMap[i, j] <= 31:
            pixelNew[i, j] = 0
        elif 32 <= pixelMap[i, j] <= 63:
            pixelNew[i, j] = 31
        elif 64 <= pixelMap[i, j] <= 95:
            pixelNew[i, j] = 63
        elif 96 <= pixelMap[i, j] <= 127:
            pixelNew[i, j] = 95
        elif 128 <= pixelMap[i, j] <= 159:
            pixelNew[i, j] = 127
        elif 160 <= pixelMap[i, j] <= 191:
            pixelNew[i, j] = 159
        elif 192 <= pixelMap[i, j] <= 223:
            pixelNew[i, j] = 191
        elif 224 <= pixelMap[i, j] <= 255:
            pixelNew[i, j] = 7

img.save('lena_2.jpg')
z = numpy.asarray(Image.open('lena_2.jpg'))
```

In the code above:

- The image is loaded and converted to grayscale using .convert('L').
- The image is processed pixel by pixel. The comparison is directly made using pixelMap[i, j].
- The if-elif structure checks which of the eight 32-level bins the pixel belongs to.
- A new pixel value from 0 to 7 is assigned depending on the range the original intensity falls into.
- The quantized image is saved and loaded as a NumPy array for further processing or analysis.

This approach is a basic example of **uniform quantization**, where the intensity range (0–255) is equally divided and mapped to fewer levels.

1) What is the purpose of using .convert('L') when loading the image? *1 point*

- To convert the image to RGB
- To resize the image
- **To convert the image to grayscale**
- To apply a filter

Yes, the answer is correct.
Score: 1
Accepted Answers:
To convert the image to grayscale

2) What type of quantization is implemented in the code? *1 point*

- Non-uniform quantization
- Logarithmic quantization
- **Uniform quantization**
- Variable-level quantization

Yes, the answer is correct.
Score: 1
Accepted Answers:
Uniform quantization

3) How many quantization levels are created in the program? *1 point*

- 255
- 8
- 16
- 4

Yes, the answer is correct.
Score: 1
Accepted Answers:
8

4) Which intensity range is mapped to the quantized value 3? *1 point*

- 0–31
- 64–95
- **96–127**
- 160–191

Yes, the answer is correct.
Score: 1
Accepted Answers:
96–127

5) What is the purpose of the pixelMap = im.load() line? *1 point*

- It saves the image to disk
- It loads pixel data into a NumPy array
- It opens the image file
- **It gives access to individual pixels of the image**

Yes, the answer is correct.
Score: 1
Accepted Answers:
It gives access to individual pixels of the image

6) Which line ensures that the output image remains in grayscale mode? *1 point*

- img = Image.new(im.mode, im.size)
- **img = Image.new('L', im.size)**
- im = Image.open('images.jpg')
- J = numpy.asarray(Image.open('lena_2.jpg'))

Yes, the answer is correct.
Score: 1
Accepted Answers:
img = Image.new('L', im.size)

7) What is the role of the line img.save('lena_2.jpg')? *1 point*

- It loads a new image
- It displays the image
- **It writes the quantized image to a file**
- It converts the image to a NumPy array

Yes, the answer is correct.
Score: 1
Accepted Answers:
It writes the quantized image to a file

8) Which of the following quantized values corresponds to the intensity range 192–223? *1 point*

- 5
- **6**
- 4
- 2

Yes, the answer is correct.
Score: 1
Accepted Answers:
6

9) Why is there no separate variable like intensity used in the loop? *1 point*

- Because pixelMap[i, j] directly gives the quantized value
- **To save memory by avoiding additional variables**
- To increase the range of pixel values
- To ensure the image remains in color mode

Yes, the answer is correct.
Score: 1
Accepted Answers:
To save memory by avoiding additional variables

10) What would happen if the input image was not converted to grayscale before quantization? *1 point*

- The program would raise an error
- **The pixelMap[i, j] would return a tuple instead of a single value**
- The quantization would still work correctly
- The image would be blurred

Yes, the answer is correct.
Score: 1
Accepted Answers:
The pixelMap[i, j] would return a tuple instead of a single value

Imagine you are helping a teacher automate the task of managing student names. The teacher has stored all the names of students who registered for a workshop in a text file called students.txt. Each line in the file contains one student's name.

You've been asked to:

1. Read all the student names from the file.
2. Count how many students registered.
3. Write all the names into a new file registered.txt, prefixed by a serial number (like 1. Riya, 2. Arjun, etc.)

Here is the Python program that does this:

```
# Step 1: Open the file in read mode
file = open("students.txt", "r")

# Step 2: Read all lines from the file
lines = file.readlines()

# Step 3: Close the file after reading
file.close()

# Step 4: Count the how many students registered
count = len(lines)
print("Total students registered:", count)

# Step 5: Open a new file to write the numbered list
new_file = open("registered.txt", "w")

# Step 6: Write each name with a serial number
for i in range(count):
    name = lines[i].strip()  # remove \n
    new_file.write(f"{i+1}. {name}\n")

# Step 7: Close the output file
new_file.close()
```

11) What does the mode "r" in the line open("students.txt", "r") mean? *1 point*

- Rewrite the file
- **Read the file**
- Remove the file
- Run the file

Yes, the answer is correct.
Score: 1
Accepted Answers:
Read the file

12) What will file.readlines() return? *1 point*

- A string with the entire file content
- A dictionary of words
- **A list of lines from the file**
- Only the first line of the file

Yes, the answer is correct.
Score: 1
Accepted Answers:
A list of lines from the file

13) What is the purpose of file.close()? *1 point*

- Deletes the file
- **Saves memory and finishes file operations**
- Prints the file
- Converts the file to uppercase

Yes, the answer is correct.
Score: 1
Accepted Answers:
Saves memory and finishes file operations

14) In the line name = lines[i].strip(), what does .strip() do? *1 point*

- Removes only spaces
- Adds a newline character
- **Removes newline and extra spaces from both ends**
- Converts the name to lowercase

Yes, the answer is correct.
Score: 1
Accepted Answers:
Removes newline and extra spaces from both ends

15) What is the effect of opening a file in "w" mode? *1 point*

- It appends content to the file
- It reads content from the file
- **It overwrites the file if it already exists**
- It deletes the file

Yes, the answer is correct.
Score: 1
Accepted Answers:
It overwrites the file if it already exists

16) What will be written in registered.txt if students.txt contains 3 names? *1 point*

- Just the count of students
- **The names with numbering like 1. Name**
- Only the last student
- Nothing, since the file is empty initially

Yes, the answer is correct.
Score: 1
Accepted Answers:
The names with numbering like 1. Name

17) What is the role of \n in the line:

new_file.write(f"{i+1}. {name}\n")? *1 point*

- It creates an error
- It writes a space
- **It moves to a new line after each entry**
- It removes the last line

Yes, the answer is correct.
Score: 1
Accepted Answers:
It moves to a new line after each entry

18) Which line in the code ensures that the new file is actually created? *1 point*

- open("students.txt", "r")
- lines = file.readlines()
- **new_file = open("registered.txt", "w")**
- print(count)

Yes, the answer is correct.
Score: 1
Accepted Answers:
new_file = open("registered.txt", "w")

19) What will len(lines) return? *1 point*

- The number of characters in the file
- **The number of lines (students) in the file**
- The size of the file in bytes
- The first line of the file

Yes, the answer is correct.
Score: 1
Accepted Answers:
The number of lines (students) in the file

20) Why do we use f"{i+1}. {name}\n" instead of i + name?

- **Because numbers and strings cannot be added directly**
- Because i is a float
- Because we need to store binary values
- Because name is an integer

Yes, the answer is correct.
Score: 1
Accepted Answers:
Because numbers and strings cannot be added directly